SUPHARADA JUNDOK (李沛妤)

Undergraduate Researcher · Machine Learning · Few-Shot Learning
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Summary

Undergraduate machine learning researcher with hands-on experience in few-shot learning and UAV remote sensing at National Dong Hwa University. Co-developed and published a CAML-based pipeline for forest carbon sink estimation at ACM BDSIC 2025, earning Best Presentation Award. Seeking an internship of **Machine Learning Scientist or Machine Learning Engineer** to apply research expertise in applied ML and AI settings across industry in Asia. Also exploring AI safety and trustworthy AI as a long-term research interest.

Skills

ML & Deep Learning: Few-Shot Learning, Feature Engineering, CNN, ResNet50, EfficientNet, GANs, PyTorch, Episodic Training, Scikit-learn, XGBoost, LightGBM
Computer Vision: Image Processing, Image Classification, Feature Extraction, Segmentation, OpenCV
Data & Viz: Pandas, NumPy, Matplotlib, Seaborn, SHAP, Jupyter
Remote Sensing & GIS: QGIS, GeoTIFF, CHM Processing, UAV RGB, Rasterio
Programming & Tools: Python, C, C++, SQL, Git, GitHub, VSCode, Linux
Cybersecurity: SIEM, IDS, Network Security, Threat Analysis

Professional Experience

Undergraduate Research Assistant — Few-Shot Learning — *ICMN Lab, National Dong Hwa University* Sep 2025 – Present
Carbon Sink Estimation in Primary Forest via Tree Segmentation and Species-Level Classification— *ACM*

- Built **custom CAML few-shot learning pipeline** with ResNet50 backbone at Danongdafu Forest Park, Taiwan; achieved **70.72% accuracy** under severely limited labeled data conditions
- Achieved **92.59% carbon sink accuracy** vs. ground-truth survey — supporting Taiwan's 2025 carbon fee policy, saving over **USD 167K in monitoring costs over 30 years**
- Preprocessed GeoTIFF UAV imagery, crown annotation, and validated geospatial data using **QGIS** and **CHM analysis** for a primary forest dataset in Taiwan
- 2025 NDHU CSIE Undergraduate Project Exhibition **– Honorable Mention**
- Supharada Jundok and Wei-Che Chien (Advisor), "Carbon Sink Estimation in Primary Forests via Individualized Tree Segmentation and Species-Level Classification," *2025 7th International Conference on Big-data Service and Intelligent Computation (BDSIC 2025)*, Bangkok, Thailand, Oct. 2025. **[Best Presentation Award]**

Advisor: Prof. Wei-Che Chien · Team member: Aaron Kuo

Projects

Alzheimer's Disease Prediction Using Machine Learning — *Final Project · Big Data Analytics, National Dong Hwa University* Dec 2025
Score: 95/100 (Presentation), 100/100 (Paper)

- Evaluated 11 ML classifiers; best model (Random Forest) achieved **96.74% test accuracy** and **95.39% F1-score** with 5-fold CV
- Developed EDA-driven **feature engineering & selection** pipeline (Chi-square, t-tests, Pearson/Spearman + heatmap); reduced 35 → 5 discriminative features
- Addressed class imbalance via **scale_pos_weight**; extended with **SHAP** for model interpretability

Advisor: Prof. Chung Yung · TA: Demiah Kisla Charlery

Education

B.S. Computer Science and Information Engineering (International Program) Sep 2022 – Jul 2026 (Expected)
National Dong Hwa University, Taiwan
GPA: 3.12 · Coursework: Introductory ML, Big Data Analytics, Operating Systems, Intro to Generative AI, Project Management

Certifications

Google Cybersecurity Professional Certificate — *Coursera · Google X NTUT (國立臺北科技大學)* Jul – Dec 2024
8 cybersecurity certificates · Network security, SIEM, IDS, threat analysis, security automation with Python

Languages

Thai & Laotian: Native · **English:** Professional (TOEIC 825) · **Chinese (Traditional):** Basic (TOCFL A2) · **Spanish:** Basic